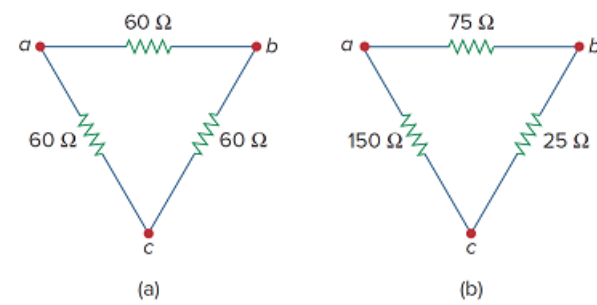


Problem

Transform the circuits in Fig. 2.113 from Δ to Y.

Figure 2.113:



Step-by-step solution

Step 1 of 5

(a)

Consider the circuit of textbook Figure 2.113(a). All the resistors are of same value. Therefore the Y circuit of Figure 2.113(a) will have resistors of same value. The Y circuit is as shown in Figure 1.

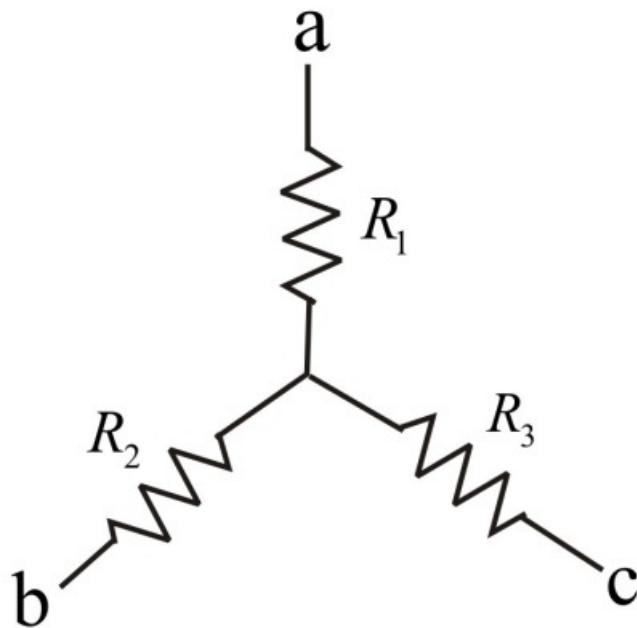


Figure 1

Step 2 of 5

Calculate the value of resistors.

$$\begin{aligned}(R_1 = R_2 = R_3) &= \frac{(60)(60)}{60 + 60 + 60} \\ &= \frac{3600}{180} \\ &= 20 \, \Omega\end{aligned}$$

Therefore, the resistors of the Υ network are $R_1 = R_2 = R_3 = 20 \, \Omega$.

Step 3 of 5

(b)

Consider the circuit of textbook Figure 2.113(b). The Υ network of the given circuit is as shown in Figure 2.

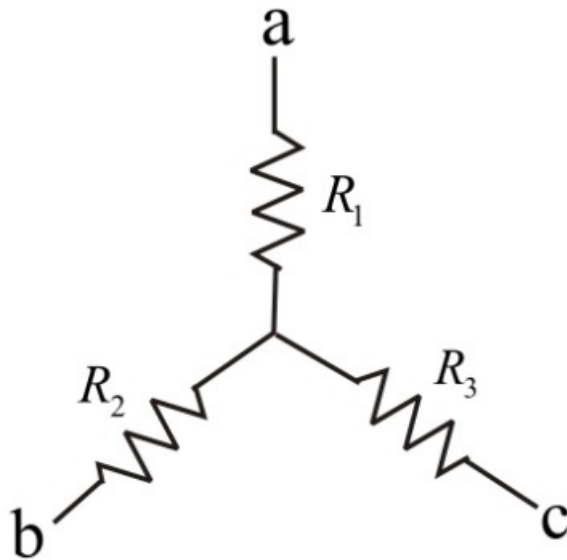


Figure 2

Step 4 of 5

Calculate the resistor value R_1 .

$$\begin{aligned}R_1 &= \frac{(75)(150)}{75 + 150 + 25} \\ &= \frac{(75)(150)}{250} \\ &= 45 \, \Omega\end{aligned}$$

Calculate the resistor R_2 .

$$\begin{aligned}R_2 &= \frac{(75)(25)}{75 + 150 + 25} \\ &= \frac{(75)(25)}{250} \\ &= 7.5 \, \Omega\end{aligned}$$

Step 5 of 5

Calculate the resistor R_3 .

$$\begin{aligned} R_3 &= \frac{(150)(25)}{75 + 150 + 25} \\ &= \frac{(150)(25)}{250} \\ &= 15 \, \Omega \end{aligned}$$

Therefore, the resistors of the ∇ network of Figure 2 are

$$R_1 = 45 \, \Omega, R_2 = 7.5 \, \Omega, R_3 = 15 \, \Omega.$$